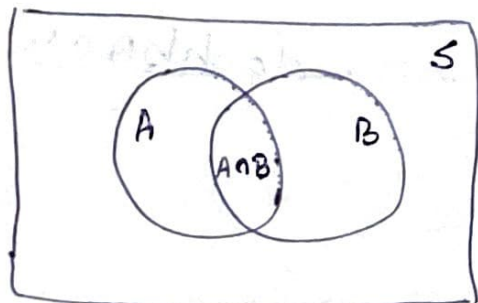


Additive Rules

- It is quite easy to calculate the probability of some event from known probabilities of other events.
- Additive rule, applies to the union of 2 events -

If A & B are 2 events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$P(A \cup B)$ is the sum of the probabilities of the sample points in $A \cup B$. When we do

$P(A) + P(B)$ we get sum of probabilities in A and probabilities of B. But, in this way, we have added the probabilities in $(A \cap B)$ twice. Therefore, we have to subtract it once.

- If A & B are mutually exclusive events,
$$P(A \cup B) = P(A) + P(B)$$

If A & B are exclusive, $A \cap B = \emptyset$, then,
 $P(A \cap B) = P(\emptyset) = 0$

- In general, $A_1, A_2, \dots, A_n$ are mutually exclusive,

$$P(A_1 \cup A_2 \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

- A collection of events $\{A_1, A_2, \dots, A_n\}$ of a sample space S is called a partition of S if $A_1, A_2, \dots, A_n$ are mutually exclusive.
& $A_1 + A_2 + \dots + A_n = S$, then

$$P(A_1 \cup A_2 \dots \cup A_n) = 1 = P(S)$$

- For 3 events, A, B & C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(B \cap C) - P(C \cap A) \\ + P(A \cap B \cap C)$$

- If events A & A' are complementary events then,

$$P(A) + P(A') = 1$$

Why,

Since $A \cup A' = S$, and the sets A & A' are disjoint.

$$1 = P(S) = P(A \cup A') = P(A) + P(A')$$